



Introduction to Pharmacology

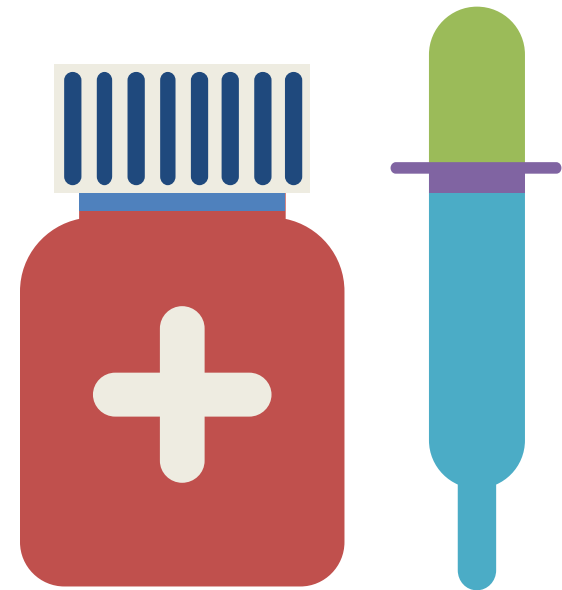
Department of Clinical Pharmacology
Faculty of Medicine,
KAU



Objectives

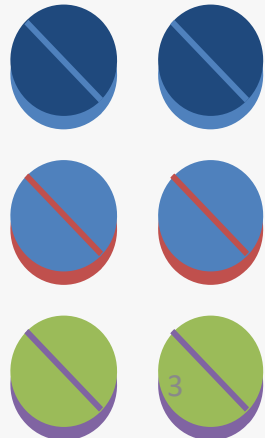
By the end of this lecture, you will be able to :

- Define key pharmacological terminologies
- Identify the goal of studying Pharmacology
- Explain the rational use of drugs
- Discuss drug characteristics : sources, classification, names
- Identify phases of drug development



Definitions

- **Pharmacology:** is the study of drugs and how they interact with living cells.
- **Medical Pharmacology:** is the science dealing with substances used to *prevent, diagnose and treat diseases*, including their pharmacokinetics, pharmacodynamics, therapeutic indications, adverse effects, dosage forms, and routes of administration.



Definitions

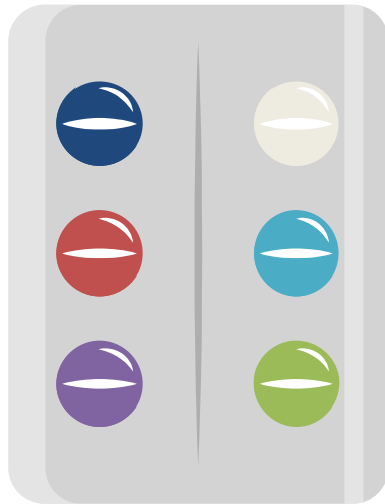


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- **Drug:** can be defined as any chemical substance that produces a change in biologic function through its chemical actions.
- Drugs may be used to promote **healing**; control or slow **disease progression**; reduce the risk of **complications**; and serves as a **replacement** therapy.

*

Major Uses of Drugs



Symptomatic

Relieve disease symptoms e.g. Aspirin

Preventative

To avoid getting a disease e.g. Hepatitis B vaccine, Flu vaccine

Diagnostic

Help determine disease presence e.g. Radioactive dyes

Curative

Eliminate the cause of the disease e.g. antibiotics

Health Maintenance

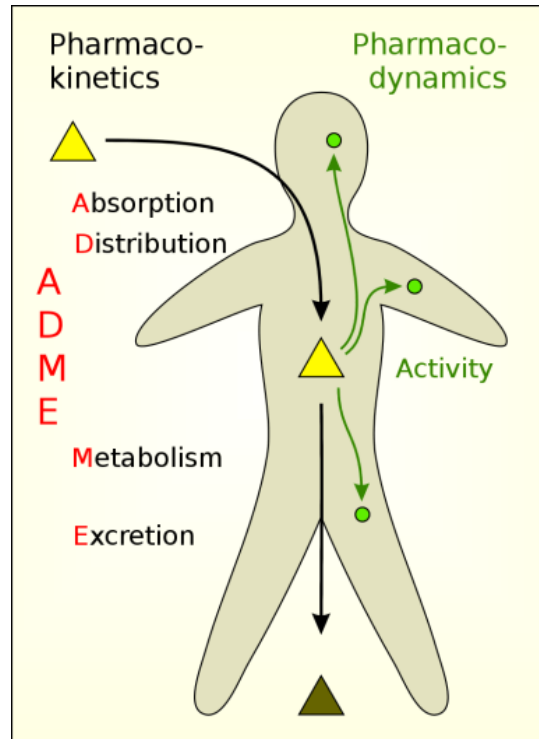
Drugs for weight control

Definitions Used in Pharmacology

Pharmacokinetics (PK)

is “what the body
does to the drug”

How the body **A**bsorb,
Distribute, **M**etabolize,
and **E**xcrete the drug.
(ADME)



Pharmacodynamics (PD)

is “what the drug
does to the body”

The study of drug
effects and their
mechanisms of action.

Definitions used in Pharmacology

- **Pharmacotherapeutics:** is the study of clinical use of drug to treat disease
- **Pharmacogenetics** is the study of how genetics alters response to drugs
- **Pharmacoepidemiology** is the study of the effect of drug on population
- **Pharmacoeconomics** is the study of the cost-effectiveness of drug therapy

Rational use of drugs

When prescribing a drug, the following should be considered:

Effectiveness

Indications

Action

Safety

Contraindications

Adverse drug reactions

Interactions with other drugs & food

Route & dosage

Others

Cost

Availability of the drug

Patient compliance

■ **All members** of health care team must exercise care to promote therapeutic effects and minimize drug induced harm

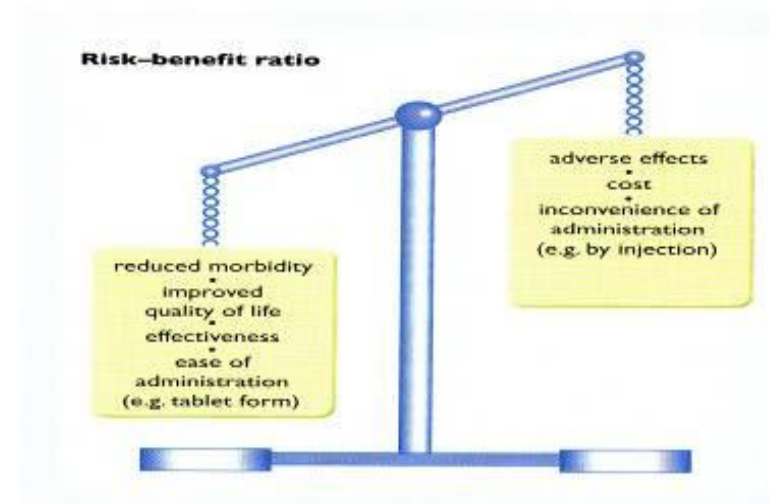
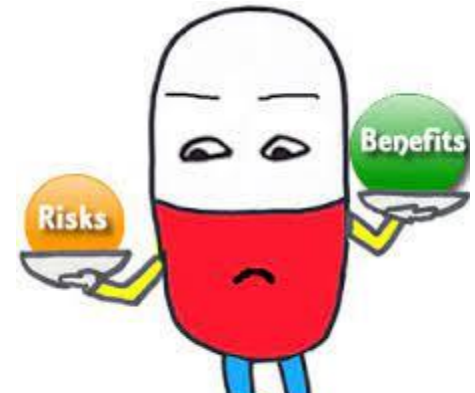


Fig. 1.3 Risk-benefit ratio. The beneficial effects of a drug should outweigh adverse effects. All drugs are capable of producing adverse as well as beneficial effects.

Risk-benefit ratio

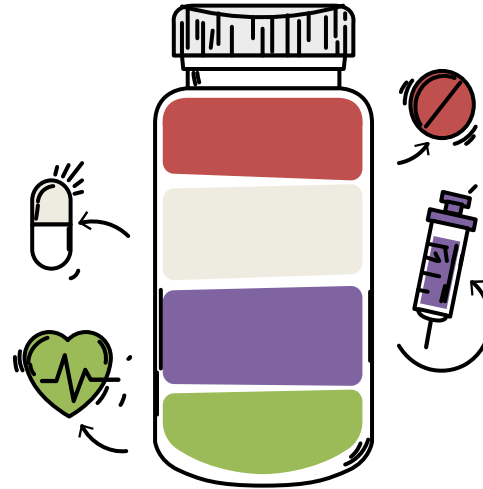
Concept 1



Almost ALL DRUGS CAN BE TOXIC

The only thing that determines if a drug provides a benefit or harm to a patient is how we administer it.

Concept 2



Therapeutic Goal

To achieve drug concentrations at the target site that produce the desired effect with no or minimal adverse reactions.

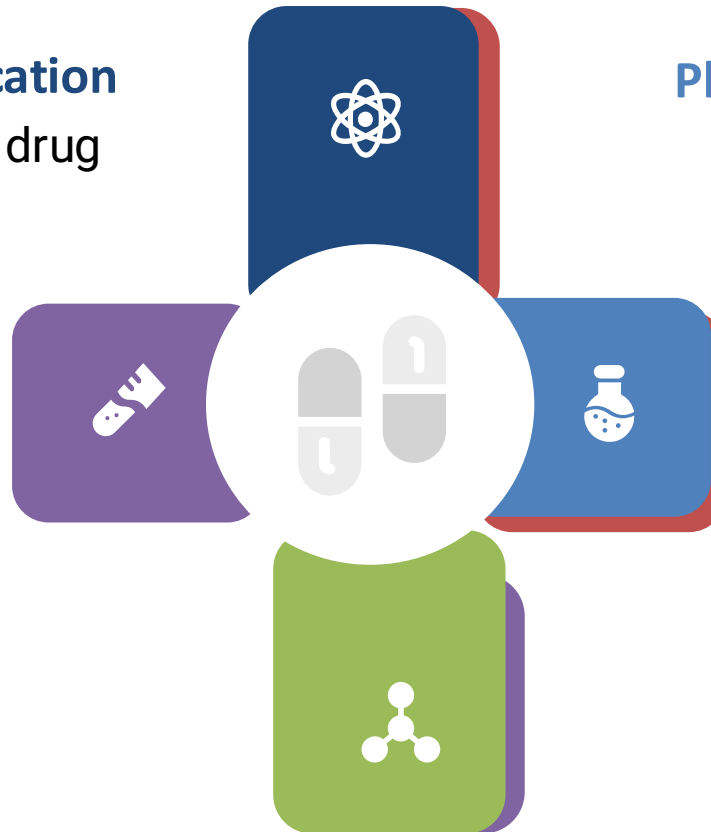
Classification of Drugs

Therapeutic Classification

Is based on what the drug does clinically

For example:

1. Anticoagulants
2. Antidepressants
3. Antineoplastics



Pharmacologic Classification

-More specific
-Based on how the drug produces its effect at molecular, tissue, or body-systems level

For example:

1. Calcium-channel blockers
2. Angiotensin-converting enzyme (ACE) inhibitors
3. Proton-pump inhibitors (PPI's)

Classification with regard to prescription

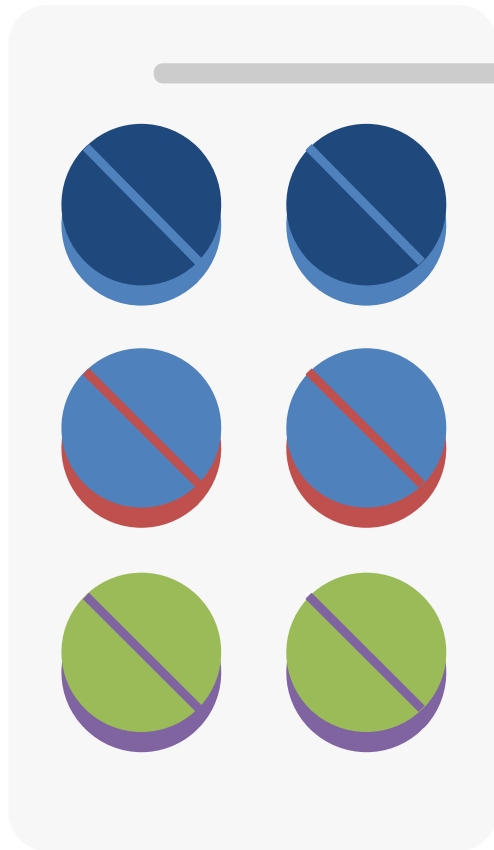
1-OTC (over the counter drugs)

2-Prescription drugs

3-Controlled substances (I-V)

I	Drugs with high abuse potential and no accepted medical use
II	Drugs with high abuse potential and accepted medical use
III	Drugs with moderate abuse potential and accepted medical use
IV	Drugs with low abuse potential and accepted medical use
V	Drugs with limited abuse potential and accepted medical use

Sources of Drugs



1

Natural

Organic or inorganic

2

Synthetic

Chemical or biological

3

Semisynthetic

I-Natural:

A- organic:

- **Plants:** alkaloids, glycosides, oils (fixed or volatile)
- **Animals:** e.g. hormones, vitamins, vaccines and heparin
- **Micro-organisms:** e.g. antibiotics as penicillin

B-Inorganic: minerals as iron, calcium & antacids.

II- Synthetic:

- ***Chemical synthesis:*** e.g. salicylates.
- ***Biological synthesis:*** by recombinant DNA technology; insertion of the desired DNA into another DNA of a living cell to form recombinant DNA e.g. human insulin.

III- Semi-synthetic:

- As morphine derivatives, and some penicillin preparations

Drug Characteristics

- **Physicochemical characteristics:**
 - ❑ **Dosage form:** Solid ; Liquid & Gaseous drugs
 - ❑ **Solubility:** Hydrophilic , lipophilic
 - ❑ **Chemical structure:** most are weak acids or bases
 - ❑ **Drug Size & Molecular weight:** Some are small simple molecules, others have complex structures , Most drugs have molecular weights between 100 and 1,000

Drug Development: From Bench to Bedside

Developing a new drug typically takes **years** (12-15) and costs **billions of dollars** to ensure acceptable **safety and efficacy**.



Clinical Trial Phases of Drug Approval

Phase I – Safety

- 20–100 healthy volunteers
- Assesses safety, tolerability, and basic pharmacokinetics

Phase II – Efficacy

- Several hundred patients with the disease
- Evaluates efficacy, dose–response, and short-term safety

Phase III – Confirmatory Trials

- Large patient populations (hundreds to thousands)
- Confirms efficacy, compares with standard therapy or placebo, and monitors common adverse effects

Phase IV – Post-Marketing Surveillance

- Conducted after approval
- Detects rare or long-term adverse effects and assesses real-world effectiveness

Differences Between Brand-Name Drugs and Their Generic Equivalents

- After the exclusive rights for a company who developed a new drug ends, competing companies can manufacture and market the generic drug for lower cost.
- Two drugs are bioequivalent if they show comparable bioavailability and similar times to achieve peak blood concentrations. Generic medications must have the same active ingredient in the same dose.
- Bioavailability is the physiological ability of the drug to reach its target cells and produce its effect.

Nomenclature (drug names)

- Chemical name
 - Acetylsalicylic acid
- Generic or official (nonproprietary) name
 - Aspirin
- Trade, brand or proprietary name
 - Jusparin, Aspocid

Nomenclature (drug names)

- Chemical name
 - *N*-acetyl-*para*-aminophenol (APAP),
Acetaminophen
- Generic or official (nonproprietary) name
 - Paracetamol
- Trade, brand or proprietary name
 - Panadol, Adol, Fefadol, Tylenol

Which name would you prescribe ?



"THE BRAND NAME IS \$57.50, AND THE GENERIC IS \$32, BUT
I CAN GIVE YOU THE PLACEBO FOR ONLY \$14.95."

Generic names give the patients more options and reduce cost

Adverse Effects

- **Augmented:** extension of pharmacological effect (dose–dependent)
- **Bizarre:** hypersensitivity and Idiosyncratic reactions (non-dose dependent).
- **Chronic:** adverse effects due to long-term use or abrupt withdrawal (e.g. corticosteroids)
- **Delayed:** Teratogenicity and carcinogenicity
- **Iatrogenic:** drug –induced disease

Pregnancy Classifications

A	No risk demonstrated to the fetus in <u>any</u> trimester
B	No adverse effects in animals— But no human studies
C	Only given after risks to the fetus are <u>considered</u> —animal studies show adverse reactions
D	<u>Definite</u> fetal risks. Only given in life threatening situations
X	Absolute fetal abnormalities & Absolute CONTRAINDICATION

Key Points:

- The **objective** of drug therapy is to achieve maximum benefit within minimal harm.
- This requires understanding drug **PK, PD** and patient-related factors.
- **Patients are unique**, drug therapy must be individualized.
- Rational use of drugs is **the responsibility** of all healthcare professionals.



Thank You

